Identifying Root Causes of Drug Addiction Using Explainable Artificial Intelligence

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***Abstract*—Drug addiction has emerged as a serious public health and social challenge in Bangladesh, particularly among young adults, driven by intertwined socio-economic, psychological, and environmental factors. Conventional statistical approaches often fail to capture the complex, non-linear interactions underlying addiction behavior, while many machine learning models operate as black boxes with limited interpretability. To address these limitations, this study proposes an Explainable Artificial Intelligence (XAI)-based framework to identify and interpret the root causes of drug addiction. Data were collected from rehabilitation centers with approval from the Department of Narcotics Control and supplemented with online survey responses. Multiple machine learning models were evaluated, where XGBoost demonstrated superior predictive performance and stability. Model decisions were interpreted using SHAP and LIME to provide both global and instance-level explanations. Grouped SHAP analysis reveals that socio-economic factors exert the strongest influence on addiction risk, followed by psychosocial determinants, while environmental factors play a moderate role. The findings demonstrate that XAI enables transparent, reliable, and policy-relevant insights that can support targeted prevention and rehabilitation strategies in Bangladesh.**

***Index Terms*—Drug Addiction, Explainable Artificial Intelligence, XAI, SHAP, LIME, Machine Learning, Public Health**

I. Introduction

Drug addiction is a chronic, relapsing brain disease characterized by compulsive drug seeking and use despite its devastating health consequences. It has evolved into a significant global public health issue, with the United Nations Office on Drugs and Crime estimating that approximately 316 million people, nearly 6 percent of the world's population aged 15 to 64 years, used illicit substances in 2023 [1], [2]. The sharp rise in substance abuse underscores the urgent need for evidence-based methodologies that can reliably identify underlying risk factors and support effective prevention strategies.

In Bangladesh, the escalation of drug addiction has reached an alarming stage. According to estimates from the Department of Narcotics Control, by 2025 nearly 8.3 million individuals may be affected by drug addiction, accounting for approximately 4.89 percent of the country's total population. A large proportion of these individuals fall within the 15 to 35 year age group, indicating an emerging youth-centered epidemic. The most commonly abused substances include cannabis, yaba, alcohol, opioids, and sedatives, reflecting easy availability and widespread misuse across diverse socio-economic groups [3].

The causes of drug addiction in Bangladesh are inherently multi-factorial, involving family conflict, unemployment, peer pressure, mental health challenges, and inadequate institutional support. These factors vary across individuals, making addiction a complex and non-linear phenomenon that cannot be fully explained using traditional analytical approaches alone [4].

Explainable Artificial Intelligence (XAI) has emerged as a promising approach to address this challenge by providing transparent and verifiable explanations behind machine learning predictions. While conventional black-box models offer strong predictive capabilities, XAI techniques such as SHAP and LIME enable stakeholders to understand which factors contribute most significantly to an individual's addiction risk and how these factors interact within the model.

This study applies XAI-based techniques to identify the underlying reasons for drug addiction in Bangladesh by integrating predictive modeling with transparent and interpretable explanations. The primary objective is to generate policy-relevant, evidence-based insights that can support early detection and effective intervention strategies within the socio-cultural context of Bangladesh.

II. Motivation

Drug addiction is a complex and multi-dimensional disorder influenced by a combination of social, psychological, economic, and environmental factors. Conventional analytical

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approaches often struggle to uncover the intricate interactions among these variables that may reveal the true underlying causes of addiction. This limitation provides strong motivation to explore advanced computational techniques capable of handling large-scale data and identifying hidden patterns that traditional methods frequently fail to detect.

From a technical perspective, modern machine learning models demonstrate impressive predictive performance; however, most of these models function as black boxes, producing predictions without explaining the reasoning behind them. In a sensitive domain such as drug addiction, this lack of transparency is a critical shortcoming, as healthcare professionals and policymakers must clearly understand why a particular prediction is made before implementing intervention strategies. Explainable Artificial Intelligence (XAI) addresses this challenge by emphasizing the interpretability of model decisions, ensuring fairness, transparency, and ethical use of data-driven systems while making AI-generated insights trustworthy for end-users.

This study is also motivated by its academic and practical significance. Through this research, valuable expertise is developed in data science, machine learning, explainable AI techniques, investigative research methodology, and ethical AI development, all of which are essential competencies in modern computer science. Moreover, working on a socially impactful problem such as drug addiction enhances professional readiness by strengthening in-demand skills relevant to healthcare analytics, public sector technology, and responsible AI-driven decision-making.

III. Objectives

The primary objectives of this study are as follows:

- To identify the major socio-economic, psychosocial, and environmental determinants of drug addiction through in-depth data analysis.
- To leverage Explainable Artificial Intelligence (XAI) techniques in order to provide transparent, interpretable, and trustworthy model decisions.
- To generate evidence-based knowledge and actionable insights for preventing and mitigating the risk of early drug addiction.

IV. Literature Review

Research on drug addiction has gradually evolved from descriptive analyses of social phenomena to more sophisticated data-driven and predictive modeling approaches. Early studies, particularly in developing countries, primarily relied on survey-based methods to examine psychosocial, familial, and environmental contributors to drug dependency. These studies consistently identified dysfunctional family structures, peer influence, cultural normalization of substance use, and socio-economic distress as major factors contributing to addiction vulnerability [4]–[8]. However, such investigations largely lacked the integration of computational intelligence or predictive modeling techniques. As a result, while they provide valuable contextual understanding, their applicability for early detection, prevention planning, and policy formulation remains limited due to the absence of predictive capability.

With recent advances in machine learning, several studies have introduced predictive models for addiction risk using demographic, psychological, and behavioral data. Many of these models have reported high predictive accuracy, with some achieving up to 97.9 percent, demonstrating the effectiveness of automated decision-support systems in addiction-related research [9]. Similarly, early risk detection frameworks in related healthcare domains have achieved performance levels exceeding 95 percent, highlighting their suitability for early warning systems [10]. Traditional parametric statistical approaches, such as logistic regression, have also been widely applied and shown success in identifying influential factors including independence from parental supervision, nightlife-related peer influence, perceived psychological stress, and treatment-seeking behavior, with reported accuracies ranging from 83 percent to 97.91 percent [2], [6], [8], [11]. Despite their strong classification performance, most of these models function as black boxes and fail to provide interpretable explanations regarding how and why specific factors influence addiction risk.

To address the limitations of black-box predictive models, recent research has increasingly explored the application of Explainable Artificial Intelligence (XAI) within e-health and substance use studies. These approaches aim to improve transparency by identifying psychological and behavioral characteristics, such as stress, impulsivity, and emotional instability, that contribute to addiction risk predictions, thereby increasing trust in machine learning outcomes [3]. However, the adoption of XAI in addiction research remains limited. Existing studies often focus on localized or feature-specific explanations rather than offering comprehensive, structured interpretations that jointly consider demographic, psychological, and socio-economic dimensions. Consequently, although predictive accuracy has improved significantly, a critical research gap persists in systematically explaining the underlying causes of drug addiction using scalable and interpretable AI methodologies.

The present study seeks to address this gap by leveraging state-of-the-art machine learning techniques alongside powerful explainable algorithms to uncover and interpret the most influential factors shaping an individual's risk profile for drug addiction. By combining predictive performance with interpretability, this work aims to contribute meaningful, evidence-based insights for addiction prevention and policy development.

V. Methodology

A. Research Design

This study adopts a quantitative, cross-sectional survey-based research design to examine the socio-economic, psychosocial, and environmental factors associated with drug addiction. As shown in Fig. 1, the framework starts with dataset acquisition, data preprocessing and target variable cleaning. A set of feature engineering methods is then employed to build up the final set of features. Both Real and Optical Tracer

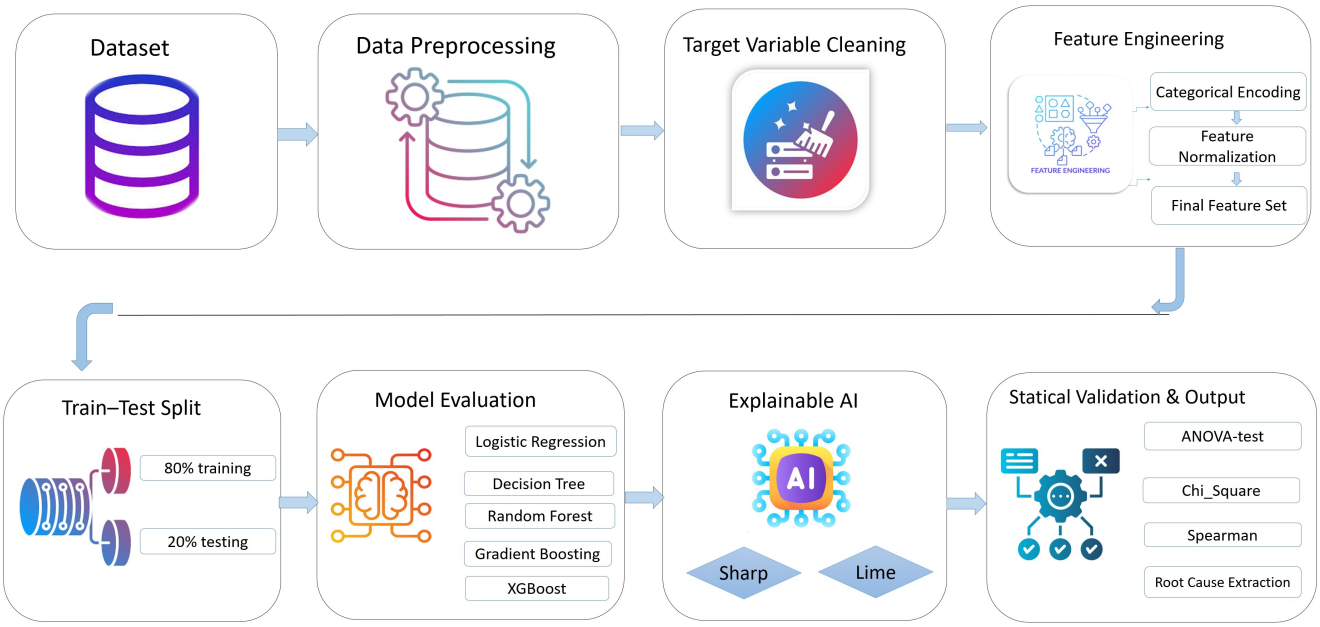


Fig. 1. Research Design Framework

The preprocessed data are separated in training and test sets over the which several machine learning models are trained and evaluated. Explainable methods (e.g., SHAP and LIME) are then used to interpret outputs of the model. And lastly, statistical validation methods are carried out to identify the robust and justifiable root causes of drug addiction.

Data were collected using a structured self-administered questionnaire designed to capture demographic information, psychological characteristics, family and social influences, substance availability, and behavioral variables related to drug use.

The study population comprised individuals aged 15 to 60 years from diverse socio-economic backgrounds residing in both urban and rural regions of Bangladesh, including both drug users and non-drug users. A purposive sampling strategy was employed to ensure adequate representation across demographic groups. Participation in the study was voluntary, and informed consent was obtained from all respondents. Participants who provided incomplete responses or withdrew during the data collection phase were excluded from the final analysis.

B. Methodological Overview

The overall research methodology follows a structured, data-driven framework aimed at identifying the root causes of drug addiction through a combination of statistical analysis, machine learning techniques, and Explainable Artificial Intelligence (XAI). The workflow is designed to ensure predictive accuracy, robustness, and interpretability of results.

The methodological procedure consists of the following steps:

C. Dataset and Study Scope

The initial dataset comprised 531 survey responses containing 19 attributes covering demographic, socio-economic, psychological, environmental, and socio-cultural factors. For the purpose of this study, only participants who reported drug use were retained for further analysis. After this filtering step, 362 confirmed drug-user records were included as the final dataset for model development and analysis.

D. Data Collection and Pre-processing

The dataset was collected through structured surveys and records obtained from rehabilitation centers. These complementary sources ensured comprehensive coverage of demographic, socio-economic, psychological, environmental, and socio-cultural characteristics related to drug use.

After data collection, the dataset was systematically cleaned and preprocessed. Inconsistent entries and noise were removed, and the dataset was prepared in a structured format suitable for subsequent statistical analysis, machine learning modeling, and explainable artificial intelligence techniques.

E. Data Preparation

The dataset included demographic, behavioral, psychological, environmental, and socio-cultural attributes. Rather than

discarding records with missing values, missing numerical features were imputed using the median, while categorical features were imputed using the mode. The dependent variable, representing the motivation or reason for drug use, was carefully cleaned and standardized. All categorical predictors were transformed into numerical form using one-hot encoding, resulting in a total of 49 binary input features.

F. Train–Test Split and Class Imbalance Handling

To ensure reliable model evaluation, the dataset was divided into training and testing subsets using an 80:20 stratified split, preserving the original class distribution. Moderate class imbalance among addiction-reason categories was addressed through the application of class weighting techniques during model training.

G. Statistical Analysis

Inferential statistical analyses were conducted to examine relationships among variables. Chi-square tests were applied to assess associations between categorical variables, while independent t-tests were used for group comparisons. Pearson and Spearman correlation analyses were performed to evaluate linear and monotonic relationships among numerical features. At a significance level of $p < 0.05$, peer pressure, psychological stress, family problems, love failure, and economic pressure emerged as statistically significant contributing factors, apart from curiosity-driven behavior.

H. Feature Importance Analysis

Feature relevance was examined using multiple complementary techniques. Pearson correlation analysis was employed to identify linear relationships between variables, while Mutual Information was used to capture non-linear dependencies. Multicollinearity and feature stability were assessed using Variance Inflation Factor (VIF) analysis to ensure robustness and reliability of model estimation.

I. Model Development

Several machine learning models were developed to predict the reasons for drug addiction, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. All models were trained under identical experimental conditions to ensure a fair and unbiased performance comparison.

J. Model Evaluation

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and ROC–AUC. Stratified validation and stability testing were conducted to assess generalization performance. Among all evaluated models, XGBoost demonstrated the most consistent and superior performance, and was therefore selected for further explainable analysis.

K. Explainable Artificial Intelligence Framework

Explainable Artificial Intelligence techniques were integrated to enhance model transparency. SHAP was employed to perform global and class-wise contribution analysis, while LIME [3] was used to generate instance-level explanations. Permutation importance analysis was additionally applied to confirm the stability and relative importance of individual features.

L. Root Cause Extraction

The final identification of root causes was achieved by integrating evidence from multiple analytical perspectives, including SHAP rankings, Mutual Information scores, permutation importance analysis, and statistical significance testing. This multi-level validation framework ensured robust, interpretable, and evidence-based identification of key risk factors associated with drug addiction.

VI. Demographic and Socio-Economic Characteristics

1) Age Distribution of the Respondents: The age distribution of the study participants is illustrated in Fig. 2. The distribution shows a higher concentration of respondents within younger age groups. The mean age of the respondents is 25.7 years, indicating that early adulthood represents a critical period of vulnerability to drug addiction. Older age groups are comparatively underrepresented in the dataset.

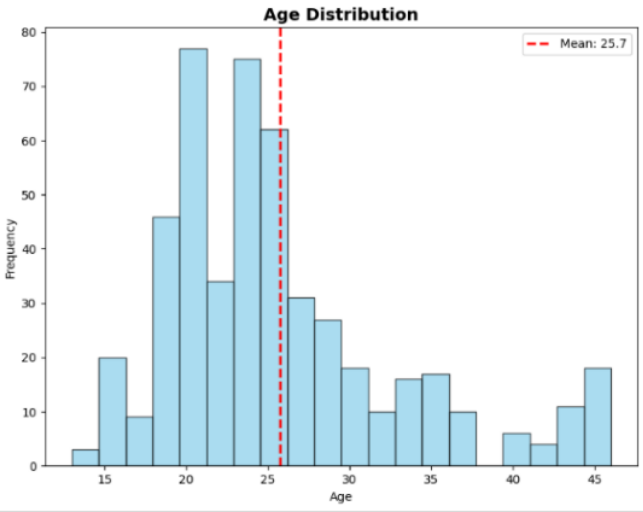


Fig. 2. Age Distribution of the Respondents

2) Gender Distribution of the Respondents: The gender-wise distribution of the respondents is presented in Fig. 3. The dataset is predominantly male, with males accounting for the majority of drug addiction cases. Female respondents represent a much smaller proportion, while participation from other gender categories is negligible. This imbalance may reflect differences in exposure, reporting behavior, and risk-taking tendencies.

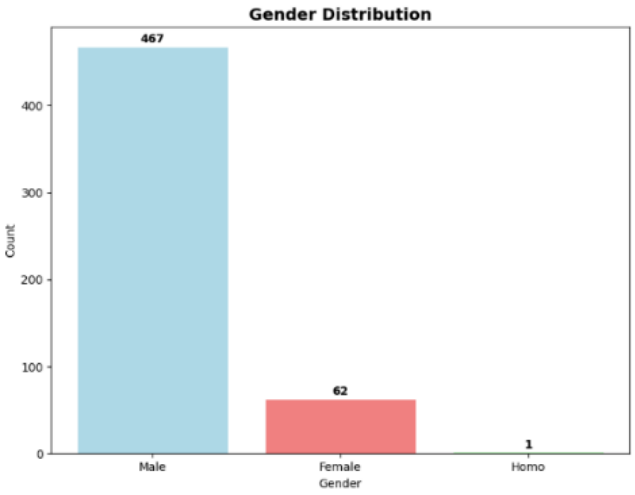


Fig. 3. Gender Distribution of the Respondents

3) Location Distribution of the Respondents: Figure 4 illustrates the residential distribution of the respondents. The results show a relatively balanced representation between urban and rural areas, with a slightly higher proportion of respondents residing in urban locations. This indicates that drug addiction affects both urban and rural populations.



Fig. 4. Location Distribution of the Respondents

4) Distribution of Employment Types: The employment status of the respondents is shown in Fig. 5. Students form the largest group in the dataset, followed by employed and unemployed individuals. Smaller proportions are observed among business owners and professionals. The dominance of students highlights the vulnerability of younger and academically engaged populations to drug addiction.

5) Drug Use Distribution: The distribution of drug use status among the respondents is depicted in Fig. 6. Drug use was reported by 68.3 percent of the participants, while 31.7 percent identified as non-users. This imbalance indicates that the dataset is predominantly composed of drug-experienced individuals, making it suitable for analyzing addiction-related risk factors. The inclusion of non-users allows for comparative and contextual analysis.

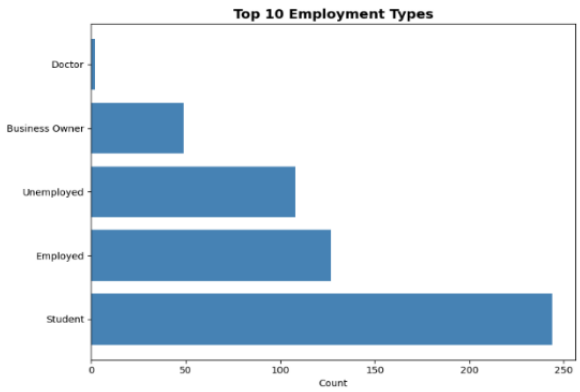


Fig. 5. Distribution of Employment Types

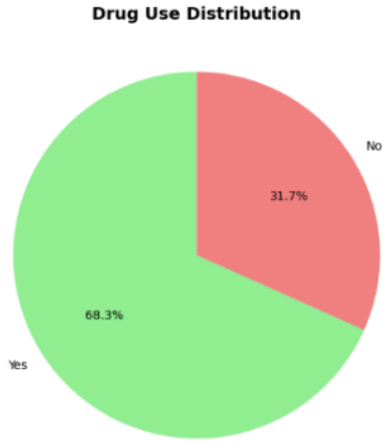


Fig. 6. Drug Use Distribution

VII. Dataset Configuration and Class Distribution

After the initial data filtering process, only respondents who reported drug use were retained for further analysis. As a result, the final dataset consists of 362 confirmed drug-user cases, which were used for model training and evaluation. A summary of the dataset configuration is presented in Table I.

The original dataset contained 17 features encompassing demographic, socio-economic, psychological, and environmental attributes. Following preprocessing and categorical encoding, these features were transformed into 49 binary encoded variables. The target variable represents the primary reason for drug addiction and contains 10 distinct classes, indicating that the problem is multi-class in nature.

To ensure robust model evaluation, the dataset was divided into training and testing subsets using an 80:20 split. This strategy provided sufficient data for model learning while preserving an independent test set for unbiased performance assessment. The overall dataset characteristics and class distribution are illustrated in Fig. 4.3.3.

TABLE I
DATASET DESCRIPTION

Item	Description
Total samples (after filtering)	362
Total features (before encoding)	17
Encoded features	49
Target variable	Reason for drug addiction
Number of target classes	10
Train–test split	80% training, 20% testing

VIII. TESTING, EVALUATION, AND COMPARATIVE ANALYSIS

A. Spearman Correlation Analysis

The Pearson correlation heatmap illustrates the linear relationships among numerical variables across demographic, psychological, emotional, financial, and socio-cultural attributes, as shown in Fig. 7. A strong positive association is observed between Age and Start Age ($r = 0.68$), indicating that individuals who are older at present generally initiated drug use later in life. Psychological stress exhibits moderate positive correlations with Life Failure ($r = 0.57$), Love Failure ($r = 0.46$), and Financial Pressure ($r = 0.44$), demonstrating the combined influence of emotional distress and economic pressure. Financial Pressure also shows moderate associations

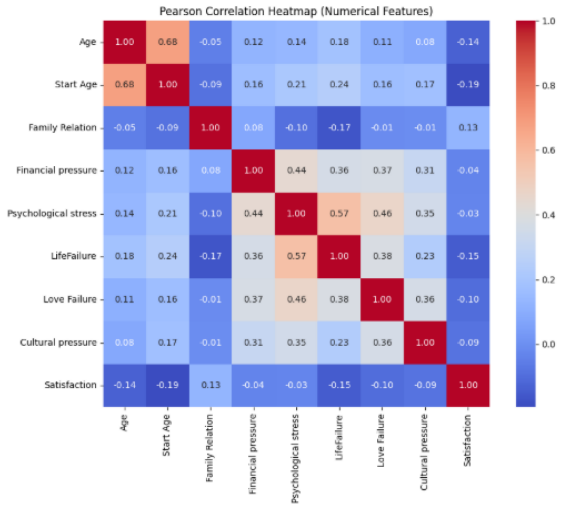


Fig. 7. Correlation Heatmap of Numerical Features

with Love Failure and Cultural Pressure, reflecting the interaction between socio-economic and emotional challenges. In contrast, Family Relation and Satisfaction display weak correlations with most variables, suggesting indirect or non-linear effects. Overall, the lack of strong linear correlations among most features justifies the use of non-linear machine learning and explainable artificial intelligence methods.

B. Mutual Information Scores

Mutual Information (MI) scores capture both linear and non-linear dependencies between predictors and the target variable. As illustrated in Fig. 8, Age and Start Age achieve the highest MI values, indicating that age-related factors are the most informative predictors of drug addiction reasons. Psychosocial attributes such as Life Failure, Family Relation, Cultural Pressure, and Psychological Stress also show relatively high MI scores, highlighting their importance. Several fine-grained

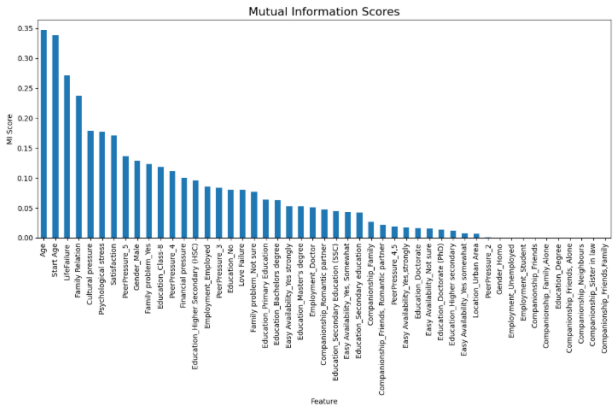


Fig. 8. Mutual Information Scores of Drug Addiction Prediction

categorical variables demonstrate low MI values, implying indirect contributions. Collectively, these findings confirm the non-linear complexity of addiction behavior and support the adoption of non-linear models and explainable AI techniques.

C. Model Performance Comparison

The predictive performance of all trained machine learning models is summarized in Table II. The evaluation metrics include accuracy, weighted precision, recall, and F1-score. Among all models, XGBoost achieved the best overall performance with an accuracy of 0.7945 and an F1-score of 0.7817, demonstrating strong multi-class classification capability. Random Forest and Decision Tree also performed well, though slightly below XGBoost, while Gradient Boosting showed moderate performance. Logistic Regression yielded the weakest results, highlighting the limitations of linear models in handling complex addiction-related data.

TABLE II
PERFORMANCE SUMMARY OF ALL MACHINE LEARNING MODELS

Model	Accuracy	Precision	Recall	F1-score
XGBoost	0.7945	0.7908	0.7945	0.7817
Random Forest	0.7671	0.7540	0.7671	0.7427
Decision Tree	0.7534	0.7331	0.7534	0.7307
Gradient Boosting	0.7397	0.7441	0.7397	0.7262
Logistic Regression	0.5342	0.5528	0.5342	0.5185

D. Confusion Matrix Analysis (XGBoost)

Figure 9 presents the confusion matrix of the XGBoost model. High true positive values along the diagonal indicate accurate classification for dominant classes such as Peer Pressure, Curiosity, Failure in Love, Lack of Awareness, and Loneliness or Depression. Minor misclassifications occur

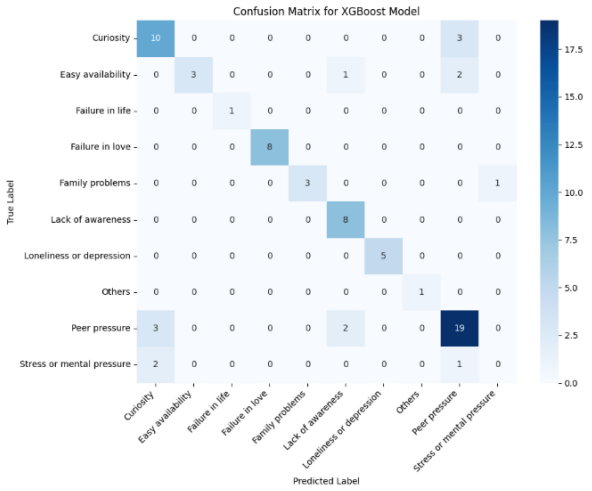


Fig. 9. Confusion Matrix Analysis of the XGBoost Model

mainly among socially and psychologically related categories, reflecting real-world overlaps between addiction drivers. Overall, the confusion matrix confirms balanced and reliable multi-class performance.

E. ROC Curve Analysis

Figure 10 illustrates the macro-average ROC curves for all evaluated models. The diagonal dashed line represents a random classifier with an AUC of 0.50. Random Forest

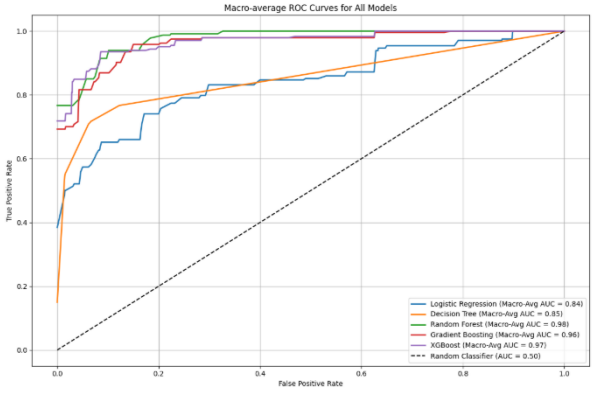


Fig. 10. Macro-average ROC Curves for All Models

achieved the highest macro-average AUC of 0.98, followed by XGBoost (0.97) and Gradient Boosting (0.96), indicating strong discriminative ability. Decision Tree and Logistic Regression exhibited lower AUC values, confirming weaker class separation. These results highlight the superiority of ensemble-based models.

F. Cross-Validation Performance Analysis

Table III reports the results of 10-fold stratified cross-validation. Random Forest achieved the highest mean accuracy (0.8094) and F1-score (0.7940) with low variance, indicating strong generalization. XGBoost and Gradient Boosting also demonstrated stable performance, while Decision Tree showed higher variance and Logistic Regression consistently underperformed.

TABLE III
CROSS-VALIDATION PERFORMANCE

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.8094 ± 0.0165	0.8204 ± 0.0330	0.8094 ± 0.0165	0.7940 ± 0.0192
XGBoost	0.7791 ± 0.0317	0.7726 ± 0.0486	0.7791 ± 0.0317	0.7652 ± 0.0338
Gradient Boosting	0.7708 ± 0.0180	0.7699 ± 0.0348	0.7708 ± 0.0180	0.7596 ± 0.0189
Decision Tree	0.7597 ± 0.0416	0.7807 ± 0.0519	0.7597 ± 0.0416	0.7548 ± 0.0440
Logistic Regression	0.5193 ± 0.0352	0.5130 ± 0.0360	0.5193 ± 0.0352	0.4944 ± 0.0387

G. Model Stability Across Random States

The stability of the XGBoost model across multiple random states is summarized in Table IV. The results show that performance remains consistent across different random initializations, with the highest accuracy observed at random states 10 and 40.

TABLE IV
MODEL STABILITY TEST ACROSS RANDOM STATES

Random State	Accuracy	Macro F1-score
10	0.7945	0.7891
20	0.7397	0.7318
30	0.7123	0.7108
40	0.7945	0.7841
50	0.7671	0.7676

IX. STATISTICAL VALIDATION AND ROOT CAUSE ANALYSIS

This section presents a comprehensive statistical validation of the relationships between input features and the target variable (reason for drug addiction). Multiple hypothesis testing methods, including ANOVA, Chi-square tests, Pearson correlation, and Spearman correlation, were applied to confirm the significance, strength, and nature of these relationships. Finally, SHAP-based analysis was used to extract and rank the root causes of drug addiction.

A. ANOVA Test Results (Numerical Features vs. Reason)

One-way ANOVA was conducted to examine whether numerical features differ significantly across addiction reason classes. As summarized in Table V, most numerical variables show statistically significant differences ($p < 0.05$), including Age, Start Age, Family Relation, Financial Pressure, Psychological Stress, Life Failure, Love Failure, and Cultural Pressure. These results indicate that the mean values of these features vary significantly across addiction categories. Satisfaction, however, does not show significant variation, suggesting limited discriminative power across addiction reasons.

TABLE V
ANOVA TEST RESULTS (NUMERICAL FEATURES VS. REASON)

Feature	F-Statistic	p-value	Significant
Age	2.6991	0.004763	Yes
Start Age	4.0159	0.000066	Yes
Family Relation	6.3060	0.000000	Yes
Financial Pressure	2.1842	0.022627	Yes
Psychological Stress	3.3893	0.000525	Yes
Life Failure	5.5939	0.000000	Yes
Love Failure	4.7486	0.000006	Yes
Cultural Pressure	3.5529	0.000307	Yes
Satisfaction	1.1591	0.320468	No

B. Chi-square Test Results (Categorical Features vs. Reason)

Chi-square tests were applied to examine associations between categorical variables and addiction reasons. Table VI shows that Education, Employment, Companionship, Family Problem, Peer Pressure, Easy Availability, and Category are significantly associated with addiction reasons ($p < 0.05$). Gender and Location do not exhibit statistically significant associations, suggesting that addiction motivations are more strongly driven by social, behavioral, and environmental contexts than by demographic attributes alone.

TABLE VI
CHI-SQUARE TEST RESULTS

Feature	Chi-square	p-value	Significant
Gender	12.9892	0.792213	No
Employment	54.2948	0.025758	Yes
Education	444.5624	0.000000	Yes
Location	15.4711	0.078783	No
Companionship	172.9729	0.000000	Yes
Family Problem	63.8487	0.000000	Yes
Peer Pressure	182.7630	0.000000	Yes
Easy Availability	100.0914	0.000005	Yes
Category	1424.0000	0.000000	Yes

C. Pearson Correlation Results

Pearson correlation analysis was conducted between numerical variables and the encoded addiction reason.

TABLE VII
PEARSON CORRELATION RESULTS

Feature	Pearson r	p-value	Significant
Age	-0.0913	0.082948	No
Start Age	-0.0170	0.747023	No
Family Relation	0.0874	0.096717	No
Financial Pressure	0.0642	0.223129	No
Psychological Stress	-0.0803	0.127038	No
Life Failure	0.0326	0.536232	No
Love Failure	-0.0873	0.097323	No
Cultural Pressure	-0.0509	0.333729	No
Satisfaction	-0.0359	0.495359	No

As shown in Table VII, all correlation coefficients are weak and statistically non-significant ($p > 0.05$). These results indicate that addiction reasons cannot be adequately explained using linear relationships, thereby justifying the use of non-linear machine learning approaches.

D. Spearman Correlation Results

Spearman rank correlation analysis further confirms weak and non-significant monotonic relationships between numerical features and addiction reasons, as shown in Table VIII. These findings highlight the limitations of rank-based statistical methods and emphasize the necessity of non-linear and explainable AI frameworks.

TABLE VIII
SPEARMAN CORRELATION RESULTS

Feature	Spearman ρ	p-value	Significant
Age	-0.0918	0.081018	No
Start Age	0.0465	0.377290	No
Family Relation	0.0533	0.311462	No
Financial Pressure	0.0636	0.227024	No
Psychological Stress	-0.0767	0.145385	No
Life Failure	0.0692	0.188835	No
Love Failure	-0.0727	0.167633	No
Cultural Pressure	-0.0631	0.231228	No
Satisfaction	-0.0435	0.408814	No

E. SHAP-based Root Cause Extraction

To identify the most influential factors driving drug addiction, SHAP-based feature importance analysis was performed. Table IX presents the ranked root causes based on mean SHAP impact. Start Age emerges as the most influential factor, followed by Age, Life Failure, and Love Failure, highlighting the importance of early exposure and emotional trauma. Socio-cultural and relational factors such as Cultural Pressure and Family Relation also play significant roles, while Psychological Stress, Peer Pressure, and Financial Pressure further reinforce the complex socio-psychological nature of addiction.

TABLE IX
SHAP-BASED IDENTIFICATION OF ROOT CAUSES

Rank	Feature	Mean SHAP Impact	Impact Direction
1	Start Age	-0.320952	Decreases Risk
2	Age	-0.213168	Decreases Risk
3	Life Failure	-0.164491	Decreases Risk
4	Love Failure	-0.145759	Decreases Risk
5	Cultural Pressure	-0.126733	Decreases Risk
6	Family Relation	-0.094716	Decreases Risk
7	Psychological Stress	-0.093863	Decreases Risk
8	Peer Pressure (Level 5)	-0.078441	Decreases Risk
9	Financial Pressure	-0.073146	Decreases Risk

Overall, the statistical and SHAP-based analyses consistently demonstrate that drug addiction is governed by complex, non-linear interactions among age-related, emotional, psychological, socio-cultural, and economic factors. These findings strongly support the use of ensemble-based machine learning and explainable AI techniques for robust and interpretable identification of addiction root causes.

X. Results and Discussion

This section provides a detailed interpretation of the experimental results obtained from machine learning models, explainable artificial intelligence (XAI) techniques, and statistical validation. The discussion focuses on predictive performance, robustness, and interpretability, with particular emphasis on identifying the root causes of drug addiction. Comparative analysis demonstrates that ensemble-based models outperform linear approaches, while XAI techniques offer transparent and clinically meaningful explanations of model behavior.

A. Distribution of Drug Addiction Reasons

The frequency distribution of drug addiction reasons after data cleaning is shown in Fig. 4.3.2. Peer pressure emerges as the most dominant reason for drug addiction, followed by curiosity. Emotional and awareness-related causes such as failure in love and lack of awareness also constitute a substantial portion of the dataset. Stress-related factors, failure in life, and other causes appear less frequently. This imbalance reflects real-world behavioral patterns and was carefully considered during model training.

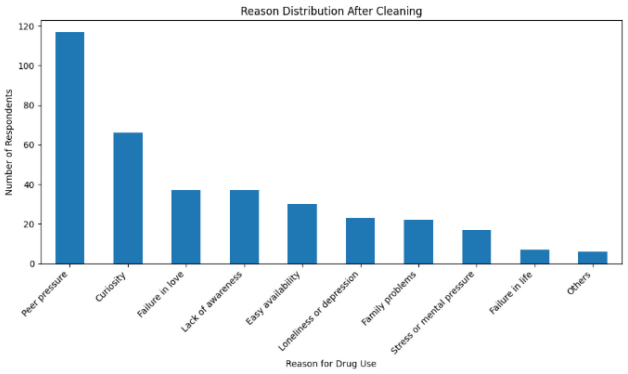


Fig. 11. Distribution of Drug Addiction Reasons After Cleaning

B. Selection of the Best Model for Explainable AI

Choosing an appropriate model for XAI requires balancing accuracy, stability, and interpretability. Linear models demonstrated limited predictive capability, whereas tree-based ensemble models performed significantly better. Among all models, XGBoost consistently achieved high accuracy, balanced F1-scores, and stable performance across different random states. Moreover, XGBoost supports SHAP and LIME-based explanations, making it suitable for both global and local interpretability. Therefore, XGBoost was selected as the final model for explainable analysis.

C. SHAP Global Feature Importance

Global feature importance was analyzed using mean absolute SHAP values. Figure 4.3.4 illustrates the overall contribution of features across all classes. Start Age is identified as the most influential predictor, followed by Age, Love Failure, and Life Failure, emphasizing the role of early initiation, life stage, and emotional stress. Socio-environmental variables such as Cultural Pressure, Financial Pressure, and Family Relation also show consistent influence, highlighting the importance of social and family context.

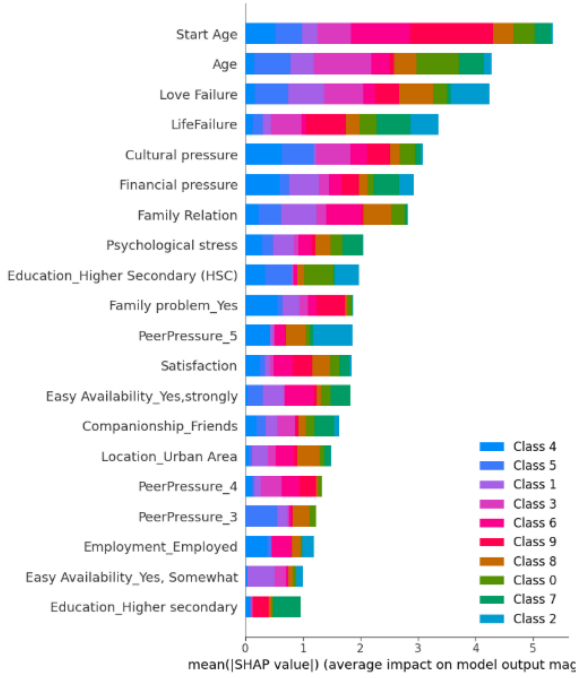


Fig. 12. SHAP Global Feature Importance

D. SHAP Dependence Plot Interpretation

SHAP dependence plots were used to explore non-linear relationships for the curiosity-driven addiction class. As shown in Fig. 4.3.5, Start Age and Age exhibit non-linear effects,

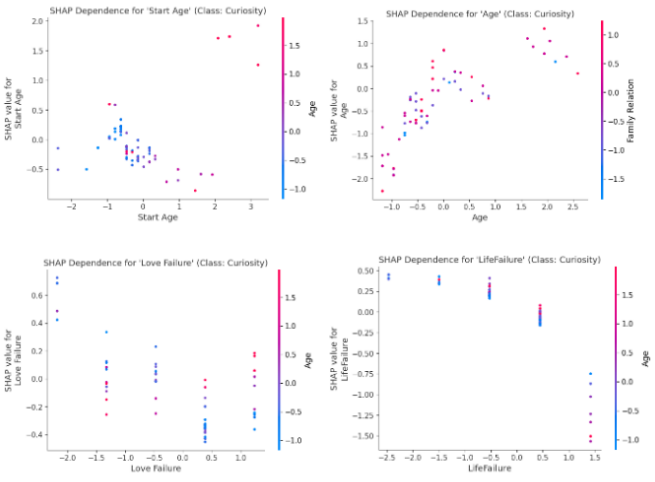


Fig. 13. SHAP Dependence Plots for Curiosity Class

where higher values increase the likelihood of curiosity-driven addiction.

Love Failure demonstrates mixed effects, generally contributing negatively at extreme levels, while Life Failure shows stronger negative influence at higher values. These patterns suggest that curiosity-driven addiction is more closely linked to exploratory behavior than severe emotional distress.

E. Class-wise SHAP Feature Importance Analysis

Class-wise SHAP analysis was performed to capture heterogeneity across addiction reasons. Figure 4.3.6 presents the top contributing features for four major classes. For Curiosity,

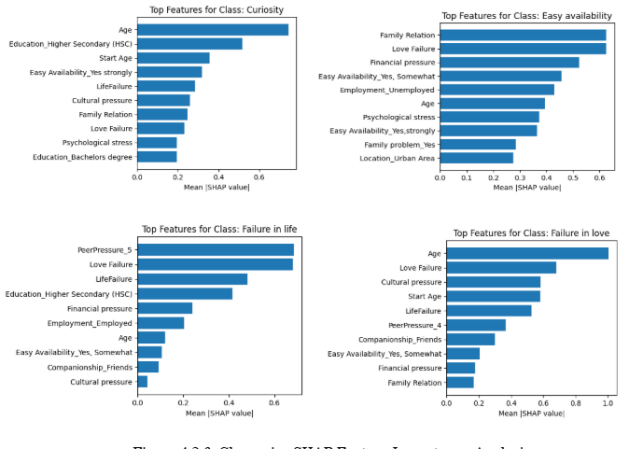


Fig. 14. Class-wise SHAP Feature Importance Analysis

age-related attributes and education dominate. In the Easy Availability class, Family Relation, Love Failure, Financial Pressure, and availability-related variables are most influential. Failure in Life is primarily driven by peer pressure, love failure, and life failure, while the Love Failure class is strongly influenced by Age, Love Failure, Cultural Pressure, and Start Age. These findings confirm that different addiction motivations arise from distinct combinations of risk factors.

F. Grouped SHAP Feature Importance Analysis

To obtain a macroscopic view, SHAP values were grouped into four categories: Socio-Economic, Psycho-Social, Environmental, and Others. Figure 4.3.9 shows that Socio-Economic factors contribute the most to addiction prediction, followed by Psycho-Social factors. Environmental factors demonstrate moderate influence, while the Other category contributes the least. This grouped analysis highlights that drug addiction is primarily driven by socio-economic and psycho-social determinants rather than environmental exposure alone.

Overall, the explainable AI analysis reveals that drug addiction is a multifactorial phenomenon characterized by complex, class-specific, and non-linear interactions among demographic, psychological, and socio-cultural factors. By combining predictive modeling with SHAP-based explanations, this study provides transparent and policy-relevant insights that can support targeted prevention and intervention strategies.

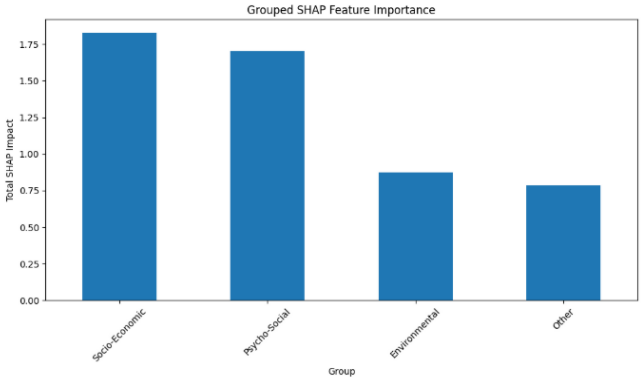


Fig. 15. Grouped SHAP Feature Importance for Drug Addiction Prediction

XI. Conclusion and Future Work

This research demonstrates that interpretable machine learning techniques can effectively uncover the underlying factors contributing to drug addiction, encompassing behavioral, psychological, social, and environmental dimensions. A comprehensive analytical pipeline was developed, including data preprocessing, categorical feature encoding, class imbalance handling, and the training of multiple machine learning models such as Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. Experimental results reveal that ensemble and boosting-based models, particularly XGBoost, exhibit superior generalization capability in predicting addiction risk, achieving the highest F1-score, ROC–AUC, and consistent performance across different random states.

Model explainability was incorporated through SHAP and LIME, enabling transparent interpretation of the most influential factors driving addiction predictions. Key contributors such as peer pressure, curiosity, family problems, emotional stress, and lack of awareness were identified as dominant determinants of drug addiction. These explainable insights not only validate the internal logic of the predictive models but also enhance transparency and trust. Such interpretability is crucial for policymakers, educators, and healthcare professionals to better understand addiction pathways and design targeted prevention and intervention strategies. Overall, the integration of machine learning with explainable AI presents a promising framework for early detection, decision support, and optimized intervention in drug abuse prevention.

Despite these contributions, the proposed framework has several limitations. First, the dataset size is relatively small and may not fully capture the diversity of addiction behaviors across different populations. Second, some variables are based on self-reported information, which may be subject to recall bias and social desirability bias, potentially affecting prediction accuracy. Third, the use of a single data source limits the generalizability of the model to broader demographic or clinical contexts. Additionally, the current approach relies on conventional machine learning models and does not incorporate temporal or real-time behavioral data, which

could further enhance predictive performance. The quality and completeness of XAI interpretations are also dependent on the availability of relevant features; omission of important psychological or environmental variables may lead to partial explanations. Finally, the proposed system has not yet undergone real-world deployment or expert clinical validation, which is essential before adoption in rehabilitation centers or large-scale prevention programs.

Future work can significantly extend both the research scope and practical applicability of this system. One important direction is the development of a full-stack web or mobile application that enables real-time risk prediction for individuals. Integrating a secure backend database such as MySQL, PostgreSQL, or MongoDB would allow efficient storage of user records, prediction history, and role-based access for counselors, educators, and administrators. Cloud-based deployment using platforms such as AWS, Azure, or Google Cloud could ensure scalability and wide accessibility.

From a research perspective, future studies should incorporate larger and more diverse datasets, including clinical assessments, socioeconomic indicators, family history, and behavioral data collected through sensors or mobile applications. Advanced deep learning or hybrid modeling approaches may be explored to capture complex dependencies in addiction behavior. Moreover, interactive dashboards with enhanced SHAP and LIME visualizations can further improve interpretability for policymakers and public health officials. Finally, collaboration with psychologists, rehabilitation specialists, and public health professionals is essential to validate the framework in real-world settings and establish it as a reliable decision-support tool for drug addiction prevention and intervention.

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